

Milad Khademi Nori

Postdoctoral Fellow, Department of Computer Science, Toronto Metropolitan University

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RESEARCH

Theory-first continual learning on foundation models. My doctoral work gave task confusion its first mathematical treatment (the Infeasibility and Feasibility Theorems, NeurIPS 2024) and produced the hybrid-replay methods that follow from it for single-machine and federated settings (ICML 2025, ICLR 2025). My current program asks why pretrained backbones evade the infeasibility result and answers with geometry: an exponential bound on task confusion in the Mahalanobis margin of the representation, a closed-form account of the bias that regularized softmax heads acquire when trained in stages, and theory-derived heads, prompts and adapters; its forward edge is memory consolidation in continual agents, formalized as an online rent-or-buy problem. **First author on every refereed paper in my record;** every result produced on a single GPU. Google Scholar: 180 citations (September 2026).

EDUCATION

Ph.D., Electrical and Computer Engineering (Machine Learning), Queen's University, Kingston 2019–2024

All degree requirements completed January 2024. Thesis: Federated Learning and Incremental Learning: Private Non-IID Learning. Supervisor: Prof. Il-Min Kim. Doctoral research funded by Huawei Technologies Canada.

M.Sc., Electrical Engineering, Amirkabir University of Technology, Tehran 2016–2019

Ranked first in the class. Thesis on energy-efficient protocols for energy-harvesting wireless sensor networks. Supervisor: Prof. Saeed Sharifian.

B.Sc., Electrical Engineering (Digital Electronic Systems), Semnan University 2012–2016

Ranked first in the class.

APPOINTMENTS

Postdoctoral Fellow, Department of Computer Science, Toronto Metropolitan University Mar 2024–present

Supervisor: Prof. Guanghui (Richard) Wang. From September 2026, a one-year research position supported jointly by Prof. Wang's group and the group of Prof. Il-Min Kim (Queen's). Two manuscripts under review at NeurIPS 2026, four at AAAI 2027, a slate prepared for ICLR 2027.

Research Assistant, Queen's University (Innovation Park), with Prof. Sidney Givigi Jan 2023–Apr 2024

Exploratory analysis of prostate-cancer datasets across modalities (CT volumes, mass spectrometry); four technical reports.

Research Assistant, Huawei-funded doctoral research program, Queen's University with Huawei Technologies Canada Aug 2019–Nov 2023

Non-IID deep learning: communication-efficient federated learning; forgetting and task confusion in continual learning. Semi-annual results presentations to Huawei; five papers in top venues.

Instructor, Lab Instructor and Teaching Assistant, ECE, Queen's University Sep 2019–Sep 2023

MATLAB programming and mathematics (about 110 students per week); discrete-time signal processing in Python (about 90 students); laboratories and mentoring.

PUBLICATIONS

Refereed (first author on all)

1. **M. Khademi Nori**, I.-M. Kim, G. Wang. Autoencoder-based hybrid replay for class-incremental learning. *ICML*, PMLR vol. 267, pp. 29753–29767, 2025.
2. **M. Khademi Nori**, I.-M. Kim, G. Wang. Federated class-incremental learning: a hybrid approach using latent exemplars and data-free techniques to address local and global forgetting. *ICLR*, pp. 84398–84411, 2025.
3. **M. Khademi Nori**, I.-M. Kim. Task confusion and catastrophic forgetting in class-incremental learning: a mathematical framework for discriminative and generative modelings. *NeurIPS*, vol. 37, pp. 47678–47707, 2024.

4. **M. Khademi Nori**, Y. Ge, I.-M. Kim. On the effectiveness of activation noise in both training and inference for generative classifiers. *IEEE Access*, vol. 11, pp. 131623–131638, 2023.
5. **M. Khademi Nori**, S. Yun, I.-M. Kim. Fast federated learning by balancing communication trade-offs. *IEEE Transactions on Communications*, vol. 69, no. 8, pp. 5168–5182, 2021. 110 citations.
6. **M. Khademi Nori**, S. Sharifian. EDMARA2: a hierarchical routing protocol for EH-WSNs. *Wireless Networks*, vol. 26, no. 6, pp. 4303–4317, 2020.
7. **M. Khademi Nori**, S. Sharifian. A QGSA cluster head selection approach for hierarchical routing protocol in the EH-WSNs. *ICSPIS*, IEEE, 2018.

Under review (first author on all, with G. Wang)

8. A Mahalanobis margin bound on task confusion in pretrained class-incremental learning: from infeasibility to exponential attenuation. *NeurIPS 2026*.
9. The structural bias of ℓ_2 -regularized cross-entropy heads in class-incremental learning: characterization, limits, and a gauge-anchoring fix. *NeurIPS 2026*.
10. Joint rehearsal restores the global gauge and leaves one scalar of bias: a closed-form theory of replay in class-incremental learning. *AAAI 2027*.
11. Two biases, one offset: long-tailed class-incremental learning in closed form. *AAAI 2027*.
12. Predicting class-incremental difficulty before training: a neural-collapse diagnostic for backbone selection. *AAAI 2027*.
13. The margin as a control variable: damped backbone fine-tuning for class-incremental learning. *AAAI 2027*.

AWARDS, SERVICE AND PRESENTATIONS

- Excellent Reviewer, IEEE Transactions on Network Science and Engineering, IEEE Communications Society, 2023. Dean's Research Fund, Travel for Postdoctoral Fellows, Faculty of Science, Toronto Metropolitan University, 2025 (ICML 2025, Vancouver). Ranked first in the class at the M.Sc. (Amirkabir, 2019) and B.Sc. (Semnan, 2016) levels. Doctoral research funded by Huawei Technologies Canada, 2019–2023.
- Reviewer for ICML, NeurIPS, ICLR, CVPR and other top-tier conferences since 2021, including ICML 2026, NeurIPS 2026 and AAAI 2027; journal reviewer for IEEE Transactions on Communications, IEEE/ACM Transactions on Networking, IEEE Transactions on Network Science and Engineering and IEEE Transactions on Pattern Analysis and Machine Intelligence.
- Presenting author at NeurIPS 2024 (Vancouver), ICLR 2025 (Singapore), ICML 2025 (Vancouver) and ICSPIS 2018; monthly seminar on continual learning in Prof. Il-Min Kim's Ubiquitous Artificial Intelligence Laboratory, Queen's University, 2019–2023; semi-annual research presentations to Huawei Technologies Canada, 2019–2023; vice-president, speaker and public-relations chair of Ingenuity Lab, Queen's University, 2019–2023, with regular talks on the impacts of artificial intelligence and mentoring of undergraduates.

SKILLS AND LANGUAGES

PyTorch, TensorFlow, NumPy, pandas, scikit-learn; timm and Hugging Face model zoos for pretrained backbones; pre-registered, seeded, single-GPU experimental practice; learning theory, margin and concentration bounds, competitive analysis of online algorithms, integer programming; Python, MATLAB, SQL, Linux, LaTeX. Persian (native), English (fluent), Arabic (reading), French (elementary).

REFERENCES

Prof. Il-Min Kim, Professor and Head, Electrical and Computer Engineering, Queen's University, ilmin.kim@queensu.ca.
 Prof. Guanghui (Richard) Wang, Department of Computer Science, Toronto Metropolitan University, wangcs@torontomu.ca.
 Prof. Sidney Givigi, Queen's University, sidney.givigi@queensu.ca. Prof. Saeed Sharifian, Amirkabir University of Technology, sharifian_s@aut.ac.ir.